rewriting the equation (8.6), we get

$$h\nu = h\nu_0 + \frac{1}{2}mv^2 \qquad (8.7)$$

The equation (8.7) is known as **Einstein's photoelectric equation**.

If the electron does not lose energy by internal collisions, then it is emitted with maximum kinetic energy K_{max}. Then

$$K_{max} = \frac{1}{2}mv_{max}^2$$

where v_{max} is the maximum velocity of the electron ejected. The equation (8.6) is rearranged as follows:

$$K_{max} = h\nu - \phi_0 \qquad (8.8)$$

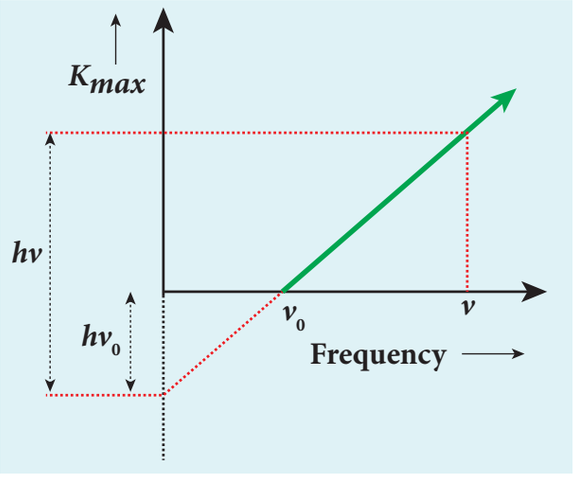


Figure 8.14 K_{max} vs ν graph

A graph between maximum kinetic energy K_{max} of the photoelectron and frequency ν of the incident light is a straight line as shown in Figure 8.14. The slope of the line is h and its y-intercept is $-\phi_0$.

Einstein's equation was experimentally verified by R.A. Millikan. He drew K_{max} versus ν graph for many metals (cesium, potassium, sodium and lithium) as shown in Figure 8.15 and found that the slope is independent of the metals.

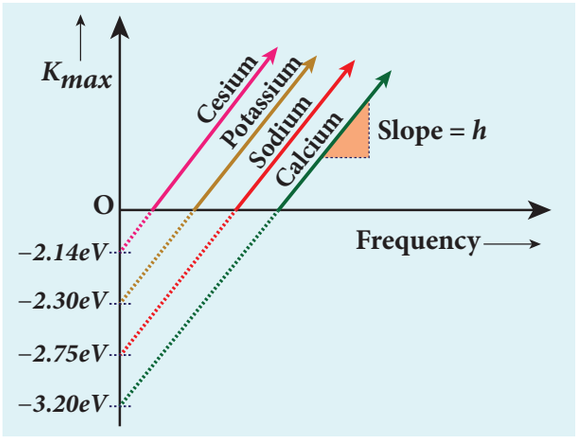


Figure 8.15 K_{max} vs ν graph for different metals

Millikan also calculated the value of Planck's constant ($h = 6.626 \times 10^{-34}$ Js) and work function of many metals (Cs, K, Na, Ca); these values are in agreement with the theoretical prediction.

Explanation for the photoelectric effect:

The experimentally observed facts of photoelectric effect can be explained with the help of Einstein's photoelectric equation.

i) As each incident photon liberates one electron, then the increase of intensity of the light (the number of photons per unit area per unit time) increases the number of electrons emitted thereby increasing the photocurrent. The same has been experimentally observed.

ii) From $K_{max} = h\nu - \phi_0$, it is evident that K_{max} is proportional to the frequency of the incident light and is independent of intensity of the light.

iii) As given in equation (8.7), there must be minimum energy (equal to the work function of the metal) for incident photons to liberate electrons from the metal surface. Below this value of energy, emission of electrons is not possible. Correspondingly, there exists minimum frequency called threshold frequency below which there is no photoelectric emission.

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